

A logarithmic-democracy obstruction for the power-profile greedy question

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Source question

The source paper is Pablo M. Berna and Hung Viet Chu, *On Some Characterizations of Greedy-type Bases*, arXiv:2201.12029. For a function $f : \mathbb{N} \rightarrow \mathbb{R}$, and a finite set $A = \{n_1 < \dots < n_m\}$, write

$$1_{f,A} = \sum_{j=1}^m f(j) e_{n_j},$$

and

$$\mathcal{D}_m^f(x) = \inf\{d(x, [1_{f,A}]) : A \subset \mathbb{N}, |A| = m\}.$$

A basis is f -greedy if there is C_f such that

$$\|x - \mathcal{G}_m(x)\| \leq C_f \mathcal{D}_m^f(x) \quad (x \in X, m \in \mathbb{N}). \quad (1)$$

The first future-research question in the source asks for a characterization of those functions f for which (1) implies ordinary greediness. The source proves this for regular f , constructs non-greedy examples for sparse f , and explicitly points to

$$f(n) = n^{-p}, \quad p > 0,$$

as a class lying outside both results.

Result

We do not settle the full power-profile problem. The following partial result shows, however, that the power profiles already force a strong quantitative form of democracy.

Theorem 1. *Let $\mathcal{B} = (e_n)$ be a semi-normalized Schauder basis of a Banach space X . Fix $p > 0$ and put $f_p(n) = n^{-p}$. If $\mathcal{B}$ is f_p -greedy with constant C_p , then $\mathcal{B}$ is suppression unconditional and there is a constant L , depending only on p , C_p , and the semi-normalization constants of $\mathcal{B}$, such that*

$$\|1_A\| \leq L \lceil \log_2(|A| + 1) \rceil \|1_B\| \quad (2)$$

whenever $A, B \subset \mathbb{N}$ are finite and $|A| = |B| \geq 1$.

Thus any counterexample to the full question for $f(n) = n^{-p}$ must be unconditional and can fail democracy by at most a logarithmic factor.

Proof intuition

The source already observes that f -greediness implies unconditionality when f has no zero values. The new point is that the power profile cannot hide large dyadic tail pieces. If a block A has at least as many earlier coordinates C before it, then

$$x = 1_{f_p, C \cup A} + \eta 1_B, \quad f_p(|C| + 1) < \eta < f_p(|C|)$$

has greedy set $C \cup B$. The greedy error is the power-weighted copy of A , while $1_{f_p, C \cup A}$ itself is an admissible one-dimensional competitor for $\mathcal{D}_m^{f_p}(x)$. Since a dyadic tail of n^{-p} has first/last ratio at most 2^p , every dyadic tail piece of A is controlled by B . Summing the dyadic pieces costs only $\log |A|$.

Proof

Let

$$a = \inf_n \|e_n\| > 0, \quad b = \sup_n \|e_n\| < \infty.$$

By Proposition 3.1 of the source paper, applied to the non-vanishing function f_p , the basis is suppression unconditional. Let K be a suppression unconditional constant. We shall only use the two standard consequences

$$\|P_E x\| \leq K \|x\|, \quad \left\| \sum_{n \in E} u_n e_n \right\| \leq K \left\| \sum_{n \in E} v_n e_n \right\| \quad \text{if } |u_n| \leq |v_n|. \quad (3)$$

Lemma 1 (tail-block estimate). *Let $C, A, B \subset \mathbb{N}$ be pairwise disjoint finite sets with*

$$|A| = |B| = r, \quad |C| = s \geq r, \quad \max C < \min A.$$

Then

$$\|1_A\| \leq C_p K 2^p \|1_B\|. \quad (4)$$

Proof. Write $C \cup A = \{d_1 < \dots < d_{s+r}\}$. Because $\max C < \min A$, the points of C are $d_1, \dots, d_s$, and the points of A are $d_{s+1}, \dots, d_{s+r}$.

Choose η with

$$f_p(s+1) < \eta < f_p(s).$$

For

$$x = 1_{f_p, C \cup A} + \eta 1_B,$$

the natural greedy set of order $s+r$ is $C \cup B$: the C -coefficients are at least $f_p(s)$, the B -coefficients are η , and the A -coefficients are at most $f_p(s+1)$. Hence

$$x - \mathcal{G}_{s+r}(x) = \sum_{j=1}^r f_p(s+j) e_{a_j},$$

where $A = \{a_1 < \dots < a_r\}$. By (3),

$$f_p(s+r) \|1_A\| \leq K \left\| \sum_{j=1}^r f_p(s+j) e_{a_j} \right\| = K \|x - \mathcal{G}_{s+r}(x)\|. \quad (5)$$

On the other hand, $1_{f_p, C \cup A}$ is an allowed competitor in $\mathcal{D}_{s+r}^{f_p}(x)$, so

$$\mathcal{D}_{s+r}^{f_p}(x) \leq \|x - 1_{f_p, C \cup A}\| = \eta \|1_B\|. \quad (6)$$

Combining (1), (5), and (6), and then letting $\eta \downarrow f_p(s+1)$, gives

$$\|1_A\| \leq C_p K \frac{f_p(s+1)}{f_p(s+r)} \|1_B\|.$$

Since $s \geq r$,

$$\frac{f_p(s+1)}{f_p(s+r)} = \left(\frac{s+r}{s+1} \right)^p \leq 2^p,$$

which proves the lemma. $\square$

Lemma 2 (disjoint logarithmic democracy). *There is a constant L_0 such that*

$$\|1_A\| \leq L_0 \lceil \log_2(|A| + 1) \rceil \|1_B\| \quad (7)$$

whenever $A \cap B = \emptyset$ and $|A| = |B| \geq 1$.

Proof. Let $m = |A| = |B|$, and write $A = \{a_1 < \dots < a_m\}$. The singleton term is controlled by

$$\|e_{a_1}\| \leq b \leq \frac{bK}{a} \|1_B\|. \quad (8)$$

For $j \geq 1$, define the dyadic tail pieces

$$A_j = \{a_{2^{j-1}+1}, \dots, a_{\min(2^j, m)}\},$$

omitting empty sets. Let $r_j = |A_j|$. The set

$$C_j = \{a_1, \dots, a_{2^{j-1}}\}$$

has cardinal $2^{j-1} \geq r_j$ and lies before A_j . Choose any $B_j \subset B$ with $|B_j| = r_j$. By Lemma 1,

$$\|1_{A_j}\| \leq C_p K 2^p \|1_{B_j}\| \leq C_p K^2 2^p \|1_B\|. \quad (9)$$

There are at most $\lceil \log_2(m+1) \rceil$ nonempty dyadic pieces, including the singleton piece accounted for in (8). Summing (8) and (9) gives (7), for example with

$$L_0 = \frac{bK}{a} + C_p K^2 2^p.$$

$\square$

We now pass from disjoint sets to arbitrary sets. Let $A, B \subset \mathbb{N}$ have $|A| = |B| = m$. Put $I = A \cap B$, $A_0 = A \setminus B$, and $B_0 = B \setminus A$. Then $|A_0| = |B_0|$, and the latter two sets are disjoint. If $A_0 = \emptyset$, there is nothing to prove. Otherwise, by suppression unconditionality and Lemma 2,

$$\begin{aligned} \|1_A\| &\leq \|1_I\| + \|1_{A_0}\| \\ &\leq K \|1_B\| + L_0 \lceil \log_2(|A_0| + 1) \rceil \|1_{B_0}\| \\ &\leq (K + KL_0 \lceil \log_2(m+1) \rceil) \|1_B\|. \end{aligned}$$

Increasing the constant gives (2). This proves Theorem 1.

Verification notes and limitations

The proof uses only the source paper’s Proposition 3.1, the definition of f -greediness, and elementary consequences of suppression unconditionality. No computational check is involved.

The result is not a full solution of the source question for $f(n) = n^{-p}$. It proves that power-profile f -greedy bases have at most logarithmic democracy distortion. The remaining frontier is whether the logarithmic loss can be removed, or whether there exists an f_p -greedy basis whose democracy constants grow logarithmically.

Novelty check

The run indexes were searched for arXiv:2201.12029, f -greedy, $\mathcal{D}_m^f$, power-profile terminology, and $1/n^p$. Only an earlier failed attempt for this same target was found. Web searches on 2026-06-21 for exact phrases around “2201.12029”, “ f -greedy”, “ $\mathcal{D}_m^f$ ”, and “ $1/n^p$ ” did not find a later resolution or a matching logarithmic-democracy statement.

References

- [1] P. M. Berna and H. V. Chu, *On Some Characterizations of Greedy-type Bases*, arXiv:2201.12029.
- [2] S. V. Konyagin and V. N. Temlyakov, A remark on greedy approximation in Banach spaces, East J. Approx. 5 (1999), 365–379.